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RaguSaucy96

...

Interesting Detail: Google Pixel 10 Pro's Main Camera has 12-bit mode enabled by default!

semiconductor.samsung.com[Open](#)

The Pixel 10 Pro XL has quietly introduced a significant upgrade to its main camera sensor, moving beyond the standard 10-bit color depth found in most smartphones, including previous Pixel models.

While the hardware itself is a familiar GNV sensor, Google has enabled its full potential by activating a secret weapon: 12-bit photo and video capture by default.

A technology from Samsung called Smart-ISO Pro, a form of Dual Conversion Gain (DCG) is actually required to achieve this, since the sensor itself is ordinarily capturing in 10-bit by default. The additional bit depth is achieved via this technology and has been previously used on older devices featuring sensors like the GN2 by using root exploits to force the mode on.

12-bit allows for greater quality in the way of smoother noise as well as improved control of the existing sensor dynamic range. This mode also inherently extends the dynamic range for a rather decent amount. Although it's

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This mode operates at the sensor level meaning that it benefits RAW image, video, and photo capture without any downsides!

How was this confirmed? Surely by now you are thinking I'm talking out of my ass and making up nonsense... Well, it's visible by running a simple Camera2API report! A kind user ([r/ausanon92](#)) was nice enough to run this report on a Pixel 10 Pro XL, showing that Google has exposed this mode as the default color depth used.

<https://www.reddit.com/r/GooglePixel/s/vRXnmWWdJB>

To see for yourself, see the Camera 2 ID which is the physical hardware ID for the main shooter lens, and find the 'white level' category. A value of 1023 corresponds to 10-bit (should be 1024 but it counts from 0 value as that's fully off) and a value of 4095 corresponds to 12-bit!

Why should you care? The latest Pixel gen is getting its cheeks clapped after all, right?

Well, this is strictly a Google pipeline issue, the actual hardware itself on main sensor is rather solid. Try out your favorite third party camera app! Blackmagic, Proshot, MotionCam - they'll all benefit from a 'mysterious' noticeably greater quality input when it comes to noise, shadows and overall dynamic range handling when compared to past pixel models!

Here's a prior showcase to demonstrate the improvements of 10-bit vs 12-bit for the curious

<https://youtu.be/e8LYd0zKMfU?si=HUCQjXc1O9WahvYe>

And yes, Google has previously stated this feature exists on conferences, their so called 'Dual Exposure' mode since the Pixel 9 release, but it's never been made available openly like this and hence gets quality gains get butchered on stock app.

This marks an awesome step forward and hopefully more OEMs start doing this. Perhaps one day the full powered DCG versions will become available on not just pixels, but other devices. Think I'm just fetishizing specs? Here's what the highest 14-bit DCG mode can do on newer sensors...

<https://www.reddit.com/r/Android/s/3vb4RVV991>

230

89

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Btw, if any P10 Pro owners want to pitch in further, a direct 10-bit vs 12-bit comparison is possible! Use the [r/MotionCamPro](#) trial and use RAW10 modes to shoot 10-bit, RAW_SENSOR to enable 12-bit! Shoot a log video sample on same settings and compare! Goes a long way 🤔

Edit: NVM! Found a volunteer!! 😄

Edit 2: Updates about the files coming in, more details on the MotionCam subreddit above 😊

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Edit 4: Here's an *amazing* video that summarizes all the developments and the end results in 7 minutes! See it all come together and how it performs in action!! <https://youtu.be/YVj6JYXF14M?si=iMzWEubj8eFygGUL>

**Quirky_Bullfrog9375** • 6mo ago

Please do give us an update, I'm planning to buy a P10P soon because I know it's reliable for my usage, love PixelOS, and the picture processing in stock Gcam. I'm also planning to buy Motioncam pro

**RaguSaucy96** OP • 6mo ago

I just posted on there - I've got hands on MCRAWs now! Check the subreddit above, I've got a crosspost of this! I'll be posting all updates there 😊

They broke something but it's definitely using a different color depth!!

**nongrata23** • 6mo ago

Chicos, :D Pixels did it, it works DCG MODE native on Pixel 10!

**cl3ft** • 5mo ago

[Post](#) Striking comparison shots in the comments.

**hectorlf** • 6mo ago

It would be awesome if Google changed the HDR implementation to this mode instead of the current approach, which constantly stores a set of frames before you press the shutter, wasting resources just by having the camera app open 🤖

Exciting times ahead!

**Legal-Branch-9318** • 6mo ago

So they don't use it ? Implement in futur software update ?

**RaguSaucy96** OP • 6mo ago

They do! It's already turned on as I mentioned. It's 'technically' in use already since the Pixel 9 if Google is to be believed but now we've got real confirmation it's the default mode, and also third

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This was not the case previously and if it's only available in stock app, they often prefer to simply use exposure bracketing for the HDR as it's in theory more effective but becomes susceptible to motion artifacts and ghosting, like staggered HDR

**Legal-Branch-9318** • 6mo ago

Thanks for precision, i've misunderstood !

**kakashi_ax** • 6mo ago • Edited 6mo ago

So we got this and a new ISP, why everything looks the same then?

**RaguSaucy96** OP • 6mo ago

I don't know, ask Google.

The device has lots of untapped potential

See for yourself what happens without the stock ISP pipeline. It's a pixel 7 Pro but the point remains...

<https://youtu.be/kxJpOqSfXp4?si=iC7vQWktXHfEGDze>

**CaptainMarder** • 6mo ago

Pixel 8,6,3,1, Nexus6p,5

why has googles camera app become so shitty, that's a drastic difference in quality.

**RaguSaucy96** OP • 6mo ago

Welcome to android 😊

join r/MotionCamPro, someone just posted a video on P7P too if you want independent showcase too. They are fvcking us with subpar ISP configurations, lol

**CaptainMarder** • 6mo ago

Pixel 8,6,3,1, Nexus6p,5

Oh is that app is designed for video? I don't know anything about videography. I have a a6400 and never use the video mode on it either lol.

But the quality difference is huge to my eyes.

**RaguSaucy96** OP • 6mo ago

Video and photography, yep! Basically shoots raw video but can shoot normal video too.

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<https://www.reddit.com/r/SonyXperia/s/xQ34LqnKKu>



CaptainMarder • 6mo ago

Pixel 8,6,3,1, Nexus6p,5

fascinating, when you say it's really intensive on storage how intensive do you mean? Like gigabytes for a few min of recording?



RaguSaucy96 OP • 6mo ago

See for yourself!

<https://play.google.com/store/apps/details?id=com.motioncam>

Trial limits you to 5 second duration per clip but otherwise what you see is what you get

Yeah, if you go raw video, quite large but it's actually data lossless compressed raw video, can be up to 50% less than typical RAW sizes based on lighting.



kakashi_ax • 6mo ago

Yeah I have motion cam and it's impressive what that app can do with the hardware available, the hardware have a lot of potential but don't know what the people at google are doing xD



JSK23 • 5mo ago

wow, what a huge improvement. google needs to be making use of this asap



RaguSaucy96 OP • 5mo ago

Ironically they actually did... *To shoot a commercial* 🎬

Check this Pixel 7 Promo https://www.youtube.com/watch?v=SyeS_xYxCLI

Pretty innocent right? Check behind scenes and comments

https://www.youtube.com/watch?v=4_GhKg4VThY

They need to really sort out the pipeline, app only uses what's already there.



GintokisRightShoe • 6mo ago

Feels like the Pixel 9 just released lol

[Skip to main content](#)[+ Create](#)**Lezlow247** • 6mo ago

My Wi-Fi and Bluetooth keep going out. My battery doesn't last as long. I can get 800 bucks for the 8 pro making it a 200 dollar upgrade. I feel like it is worth it

**GintokisRightShoe** • 6mo ago

Those are some massive issues, sounds very worth it in your case

**[deleted]** • 6mo ago**[deleted]** • 6mo ago**cbelliott** • 6mo ago

You specifically mention the XL on your post - is this only on the XL or also applies to the smaller Pro as well?

**RaguSaucy96** **OP** • 6mo ago

We don't know, but it should!!

**ceepyou** • 5mo ago

Ok, so my wife snagged a Pixel 10 in Frost (256GB) yesterday. With UFS 4.0 being the memory of choice above 128GB we knew 256GB would future proof it. It looks like Motion Cam Pro has full sensor 12-bit data on the regular P10 sensor too! I'll post screenshots and samples when I can. Maybe this little sensor will do better with the DCG help.

**RaguSaucy96** **OP** • 5mo ago • Edited 5mo ago

Well I'll be damned! It did seem to have a dcg capable sensor from what people said about it however it was unknown if they'd enable it on there too! This is an even more amazing precedent as they've done it in spite of sensor discrepancy!!

Thank you so much for reporting this! 😄

Jump onto the MotionCam Discord or [r/MotionCamPro](#) and let's push that thing!! 🙌🙌🙌🙌

Edit: Unfortunately not the case, if anyone sees this but oh well, we tried 😬

**ceepyou** • 5mo ago

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Edit: Google says USB C 3.2 but nothing else.



RaguSaucy96 OP • 5mo ago

I wouldn't worry about it, it's ample fast enough! It's got a USB 3.2 port!



ceepyou • 5mo ago

Great! If you happen to see 400-600MBps transfers that is plenty good.



RaguSaucy96 OP • 5mo ago

Hey! We actually just got hands on an API report for the P10

Question, do you see RAW12 under MotionCam streams listed? The report unfortunately doesn't show Raw12 or 4095 🤔

Could be flawed perhaps but nevertheless, Could you check if you get a chance??



ceepyou • 5mo ago

I'll take a look. I don't remember RAW12 so maybe I was too excited when I saw RAW_SENSOR.



ceepyou • 5mo ago

Just checked. Every camera is showing White Level = 1023. Sorry to cause extra excitement.



RaguSaucy96 OP • 5mo ago

Ahh darn. It's alright, at least now we know for certain 🤔

Thank you for double checking however!



pdimri • 6mo ago



MarionberryDear6170 • 6mo ago



Kortarkaick21 • 5mo ago

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are required on picture taking look after.

**RaguSaucy96** OP • 5mo ago

It's a bug that's being fixed on app side. Jump to [r/MotionCamPro](#) as it's a different topic of its own

**Tiny-Vegetable-4756** • 5mo ago

Any Possibility, Google Give it by Update in near future? Also, Android 16 is having support for APV.

<https://developer.android.com/about/versions/16/features#apv>

**quiet_brother00** • 5mo ago

Does this mean the video recording wont be choppy? So far every setting on the pixel 10 pro i've tried, except for 4k 60fps , is choppy ish.

**RaguSaucy96** OP • 5mo ago

Unrelated.

Here's a summary <https://youtu.be/YVj6JYXF14M?si=gt3L5uaSr2zvURIA>

**quiet_brother00** • 5mo ago

does this have an effect on photos as well? How does this differ from shooting photos in raw?

Also, with this app I might be able to take better photos that are jpeg as well?

**RaguSaucy96** OP • 5mo ago

It does, it's a sensor mode so impacts everything when enabled!

**Downtown-Note-3706** • 5mo ago

So is this enabled for photos only or videos also by default? I can only see 10 bit HDR on my settings? Do I need to unselect this first? Or is this mode available only from 3rd party apps?



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As stock codecs are limited to 10-bit, it would capture source at 12-bit and downscale to 10-bit which is still better.

To force mode on, yes, third party apps have that ability. This video explains it <https://youtu.be/YVj6JYXF14M?si=gnxN7jYwHuGiQqQC>

He even pits it vs other phones too <https://youtu.be/81NDEDMNAYQ?si=5ONfviXjMozh5kCU>

**Downtown-Note-3706** • 5mo ago

Is there any evidence video boost records at 12 bit and scales down to 10?

**RaguSaucy96** OP • 5mo ago • Edited 5mo ago

The evidence is empirical, logs that can be monitored while the device runs and you switch modes

You can see the logs about the requests and functions/activities triggered by the device. A vendor key and sensor mode change that are now understood to be the DCG operating mode are triggered by the device and requested only while using video boost

Users have rooted the pixel 8 and 9 to confirm, forcing the same sensor mode does indeed activate DCG on apps like MotionCam so hence we can understand it.

As the device is limited to 10-bit Codecs, it has to downscale. DCG will still yield benefits however

MotionCam uses RAW video format called MCRAW and can hold 16-bit so despite the device limitation, it can still harness the full readout therefore :)

**Downtown-Note-3706** • 5mo ago

Also, is the whole point of 12bit HDR that we have more detail to work with in post production?

**RaguSaucy96** OP • 5mo ago

12-bit gives more tonal data and also gives you a bit more dynamic range. The actual color depth $\neq$ dynamic range, however DCG gives multiple gain readouts meaning that it does increase it greatly in this case.

12-bit generally does provide better data for post editing, however DCG provides also additional denoising via merge and also mitigates shadow noise by having a dedicated exposure to it. By having dual readouts and also having double the frame readout data, the noise elimination achieved is far beyond typical denoising where you kill details; you actually enhance them!



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**Downtown-Note-3706** • 5mo ago

So is it enabled by default on video boost? What video editing software let's me edit 12bit raw vids?

**RaguSaucy96** OP • 5mo ago

So is it enabled by default on video boost?

Yes however it's limited to HEVC which gives it a max of 10-bit encoded containers.

To edit the 12-bit RAW video, you can use MotionCam app itself or any PC editor that can handle CinemaDNG.

It's more complex from here onwards so jump on subreddit or discord for more on this aspect as it's diverging from original thread 😊